Full Anti-Bluff Proofing Plan

Date: 2026-05-08 **Status:** Draft **Author:** Agent assessment after OkHttp build fix session

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Assessment: NOT Bluff-Proofed (Evidence)

Blocking (must fix before ANY release)

#	Issue	Severity	Evidence
1	`:core:auth:impl` test compilation failure	BLOCKER	`FakePreferencesStorage` does not implement `getDeviceId()`. `./gradlew test` fails.
2	All 16 submodules have uncommitted constitution propagation	BLOCKER	`git status` shows all 16 submodules/ as `-dirty` (modified CLAUDE.md/ AGENTS.md/ CONSTITUTION.md from §6.U/6.V/6.W propagation)
3	Parent repo CLAUDE.md, AGENTS.md, CONTINUATION.md have uncommitted §6.U/V/W changes	BLOCKER	Uncommitted edits sitting in working tree
4	GitFlic + GitVerse remotes still configured	§6.W VIOLATION	`git remote -v` shows `gitflic`, `gitverse`, `upstream` (gitflic) remotes
5	`origin` remote has multi-target push URLs mixing all 4 providers	§6.W VIOLATION	`origin` pushes to GitHub + GitLab + GitFlic + GitVerse simultaneously

High (features exist but verification is weak or absent)

#	Issue	Severity	Evidence
6	Challenge Tests C17-C22 never executed	HIGH	CONTINUATION.md §4.5.4 confirms they "require the §6.I emulator matrix to execute"
7	§6.I emulator matrix never run	HIGH	CONTINUATION.md §2.1 — blocked on operator hardware
8	No git tags cut for any 1.2.x release	HIGH	Last tag is Lava-Android-1.1.3-1013, current is 1.2.13-1033
9	No real-device verification evidence for 1.2.x	HIGH	CONTINUATION.md §2.1 — requires operator hands-on
10	No recent bluff hunt executed	HIGH	Last bluff hunt not documented; §6.N requires per-phase hunt
11	Many tests may be bluffs (pass while feature broken)	HIGH	Historical pattern per §6.L — "all tests green, most features broken" has happened repeatedly

Medium (known but not blocking immediate safety)

#	Issue	Severity	Evidence
12	`/api/v1/search` route resolution needs operator verification	MED	CONTINUATION.md §4.5.1
13	`/health` intermittent timeout + UDP buffer warning	MED	CONTINUATION.md §4.5.2
14	Internet Archive / gutenberg provider status unclear	MED	CONTINUATION.md §4.5.3
15	`Engine.Ktor` dead enum cascade	LOW	CONTINUATION.md §4.5.7
16	`Endpoint.Mirror` dead LAN-IP branch	LOW	CONTINUATION.md §4.5.8
17	`docs/todos/` untracked	LOW	CONTINUATION.md §4.5.9
18		LOW	

#	Issue	Severity	Evidence
	IPv4/host:port/schedule literal grep staged		CONTINUATION.md §4.5.10

Phased Plan

Phase 0: Stabilize — Fix Compilation & Commit Pending Changes

Goal: `./gradlew test` passes clean, all pending constitutional changes are committed, all 4 forbidden remotes removed.

Tasks:

#	Task	Details	Verification
0.1	Fix `:core:auth:impl` `FakePreferencesStorage.getDeviceId()`	Add the missing method override to the test fake	`./gradlew :core:auth:i` passes
0.2	Commit all 16 submodule constitution propagation	In each submodule: `git add CLAUDE.md AGENTS.md CONSTITUTION.md` (as present), commit, push to GitHub + GitLab	`git submodule status` clean, all 16 pushed mirrors
0.3	Update parent submodule pins to match committed submodule HEADs	`git add submodules/*` after submodule commits	`git diff --cached submodules/` shows n hashes
0.4	Commit parent CLAUDE.md + AGENTS.md + CONTINUATION.md §6.U/V/W changes	Pending uncommitted edits in parent working tree	`git status` clean
0.5	Remove GitFlic remote	`git remote remove gitflic`	`git remote -v` shows gitflic
0.6	Remove GitVerse remote	`git remote remove gitverse`	`git remote -v` shows gitverse
0.7	Remove upstream (GitFlic dup) remote	`git remote remove upstream`	`git remote -v` shows upstream

#	Task	Details	Verification
0.8	Fix `origin` remote to push only to GitHub + GitLab	Reset `origin` to single URL, or add separate `github/gitlab` push targets	`git remote -v` shows providers
0.9	Update scripts/tag.sh and Upstreams/ scripts to use only 2 remotes	Remove GitFlic/ GitVerse references	`grep -r 'gitflic|git` `scripts/ Upstreams/` empty
0.10	Push parent to both remotes	`git push github master && git push gitlab master`	Both converge to sa
0.11	Verify SHA convergence across both remotes	`for r in github gitlab; do echo "$r: $(git ls-remote $r master	awk '{print $1}')"; c

Exit criteria: `./gradlew test` passes. `git status` clean. `git remote -v` shows only GitHub + GitLab. All 16 submodules committed and pushed. No forbidden remotes.

Phase 1: Anti-Bluff Audit of All Existing Tests

Goal: EVERY test file verified to catch real defects — not pass while features are broken.

Tasks:

#	Task	Details	Verification
1.1	Inventory all test files	`find . -name '*Test.kt' -o -name '*Test.java*' -o -name '*_test.go'`	sort`
1.2	Random sample 5 test files, run falsifiability rehearsal (Seventh Law clause 5)	For each: (a) read the test, (b) deliberately break the production code it claims to cover, (c) confirm test FAILS with clear assertion message, (d) revert mutation, (e) document result	5 files × fail-confirmed in `.lava-ci-evidence/bluff-hunt/2026-05-08.json`
1.3			

#	Task	Details	Verification
	Check every test for §6.J clause 3 compliance (primary assertion on user-visible state)	Grep for `verify` patterns, `assertEquals` on internal state, mock assertions as primary	List of tests that need refactoring
1.4	Refactor identified bluff tests	Replace mock-only assertions with real-stack, user-visible state assertions	Bluff-Audit stamp on every refactored test commit
1.5	Check for forbidden test patterns (Seventh Law clause 4)	`mockk<SUT>` inside `*Test.kt`, `@Ignore` without tracking issue, tests that build SUT but never invoke methods	List of violations, fix or document
1.6	Verify all Challenge Tests (C1-C22) have falsifiability rehearsal protocols in their KDoc	Read each Challenge Test file	Every file has documented break-and-confirm-fail procedure in KDoc

Exit criteria: All test files audited. Minimum 5 bluff-hunt mutations confirmed to cause test failures. Zero forbidden patterns. All Challenge KDOCs document falsifiability protocol.

Phase 2: Fix All Known Issues

Goal: Every issue in §4.5 of CONTINUATION.md is either fixed (with regression test) or explicitly documented as deferred with a tracking issue.

Tasks:

#	Task	Details	Verification
2.1	Fix Challenge Tests C1-C16 if they exhibit bluff behavior	Run each Challenge, deliberately break feature, confirm failure	Each Challenge has documented falsifiability rehearsal
2.2	Execute C17-C22 on	Run each new provider Challenge Test	

#	Task	Details	Verification
	available device/emulator		All 6 pass or documented failures
2.3	Verify `/api/v1/search` route resolution	`curl -fsSk 'https://thinker.local:8443/v1/rutracker/search?q=test'`	Returns 200 with search results
2.4	Document UDP buffer fix for operator	`sudo sysctl -w net.core.rmem_max=7340032` per CONTINUATION.md §4.5.2	Operator note added to docs/
2.5	Verify Internet Archive + gutenberg provider status	Test via API call or Challenge Test	Provider works or documented limitation
2.6	Clean up `Engine.Ktor` dead enum (if touching relevant code)	Remove enum value + prune exhaustive when branches	Compiles without `Ktor` references
2.7	Clean up `Endpoint.Mirror` dead branch (if touching relevant code)	Prune dead routing branch	Compiles clean
2.8	Commit or gitignore `docs/todos/`	Operator decision	Tracked properly

Exit criteria: All HIGH and MEDIUM issues resolved or explicitly deferred with tracking. CONTINUATION.md §4.5 updated.

Phase 3: Emulator Matrix Execution (§6.I)

Goal: All Challenge Tests pass on the mandated emulator matrix inside containers.

Tasks:

#	Task	Details	Verification
3.0	Fetch latest `vasic-digital/Containers` submodule	Update pin to latest commit for emulator support	`git submodule update --init submodules/containers`
3.1	Build emulator containers for	Per Containers submodule tooling	Each container boots and shows `adb devices`

#	Task	Details	Verification
	API 28, 30, 34, latest		
3.2	Build emulator containers for phone, tablet, TV form factors	Per §6.I matrix requirements	Each form factor boots
3.3	Execute Challenge Tests C1-C22 on each matrix combination	`scripts/run-emulator-tests.sh --avds=...`	JSON evidence per combination
3.4	Document per-AVD attestation rows	Record in `.lava-ci-evidence/Lava-Android-<version>/real-device-verification.md`	Matrix coverage table complete
3.5	Falsifiability rehearsal on each AVD class	Break feature, confirm Challenge fails on each Android version class	Per-version failure evidence

Exit criteria: All 22 Challenge Tests pass on all matrix combinations. Attestation document written with per-AVD rows. Falsifiability rehearsals documented.

Phase 4: Full Bluff Hunt (Seventh Law clause 5)

Goal: Systematic verification that every test class actually catches real defects.

Tasks:

#	Task	Details	Verification
4.1	Select 5 random test files + 2 production code files from gate-shaping code	Per §6.N: sample includes `scripts/tag.sh` helpers, `scripts/check-constitution.sh`, `submodules/containers/pkg/emulator/`	Selection documented
4.2	For each file: mutate production code, confirm test fails	Follow Seventh Law clause 5 protocol	Each mutation documented in `.lava-ci-evidence/bluff-hunt/2026-05-08.json`

#	Task	Details	Verification
4.3	For any test that PASSES when production code is broken: classify as bluff, record incident	Per Seventh Law clause 6 (Bluff Discovery Protocol)	Incident recorded in `.lava-ci-evidence/sixth-law-incidents/`
4.4	Fix all discovered bluffs	Rewrite test to catch the real defect	Bluff-Audit stamp on fix commit

Exit criteria: 5+ production mutations cause test failures. Zero bluffs found, or all bluffs fixed and documented.

Phase 5: Full CI Gate & Real-Device Verification (Seventh Law clause 2-3)

Goal: Complete the pre-tag evidence pack.

Tasks:

#	Task	Details	Verification
5.0	Operator action: hardware/ device access	Connect Android device or authorize emulator matrix run	Device visible via `adb devices`
5.1	Run full CI gate	`./scripts/ci.sh --full`	All gates pass, evidence written to `.lava-ci-evidence/`
5.2	Real-device manual smoke test	Login + search + browse + download on real device against thinker.local	Recorded in real-device-attestation.json
5.3	Take 3+ screenshots or video	Document user-visible behavior per Seventh Law clause 3	Screenshots hashed and referenced
5.4	Pepper rotation	Append new pepper per `docs/RELEASE-ROTATION.md`	`.lava-ci-evidence/distribute-changelog/firebase-app-distribution/pepper-history.sha256` updated
5.5	Verify pre-tag evidence	Check all required files: `ci.sh.json`,	

#	Task	Details	Verification
	pack complete	`challenges/C{1..22}.json` at VERIFIED, `mirror-smoke/`, `bluff-audit/`, `real-device-verification.md`	`scripts/tag.sh` (dry-run) does not refuse

Exit criteria: Full evidence pack populated. Real-device verification documented.

Phase 6: Rebuild API + Client, Redistribute, Reboot

Goal: Shippable artifacts with constitutional compliance.

Tasks:

#	Task	Details	Verificatio
6.1	Rebuild Go API	`cd lava-api-go && make build`	Binary at `l go/bin/lava`
6.2	Rebuild Go API Docker image	`cd lava-api-go && make image`	Image `lava go:dev`
6.3	Rebuild Android APK (debug + release)	`./gradlew :app:assembleDebug :app:assembleRelease`	APKs in `app outputs/apk`
6.4	Rebuild proxy fat JAR	`./gradlew :proxy:buildFatJar`	JAR at `prox libs/app.ja`
6.5	Copy artifacts to `releases/`	`./build_and_release.sh`	`releases/<v` populated
6.6	Cut git tag (Android)	`scripts/tag.sh --app android`	Tag `Lava-An <vname>-<vc` pushed to l mirrors
6.7	Cut git tag (API)	`scripts/tag.sh --api go`	Tag `Lava-AP <vname>-<vc` pushed to l mirrors
6.8	Distribute via Firebase App Distribution	`scripts/firebase-distribute.sh`	New versio Firebase w changelog

#	Task	Details	Verificatio
6.9	Verify post-push SHA convergence	Check both mirrors report same tag SHA	Converged
6.10	Reboot API container	`./stop.sh && ./start.sh`	`curl https:` `thinker.loc` `health` retu
6.11	Verify API functional after reboot	Smoke test: search, auth, health	All endpoin respond co

Exit criteria: Git tags cut on both mirrors. APK on Firebase with proper versionCode and changelog. API container restarted and healthy.

Constitutional Reinforcement (if needed after audit)

After phases 1-6, if any gaps are found in the existing constitutional clauses (§6.J, §6.L, Sixth Law, Seventh Law), this section documents the amendments.

#	Clause	Current state	Amendment needed?
A	§6.J (Anti-Bluff Functional Reality)	Present in CLAUDE.md + AGENTS.md	Verify during Phase 1
B	§6.L (Operator's Standing Order)	Present in CLAUDE.md + AGENTS.md	Verify during Phase 1
C	Sixth Law (Real User Verification)	Present in CLAUDE.md + AGENTS.md	Verify during Phase 1
D	Seventh Law (Anti-Bluff Enforcement)	Present in CLAUDE.md + AGENTS.md	Verify during Phase 1

The user's mandate — "execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product" — is already codified in §6.J and §6.L. Phase 1 will determine if the mechanical enforcement is adequate.

Timeline Estimate

Phase	Effort	Dependencies	Can be parallelized?
0	~2-3 hours	None	Submodule commits can run in parallel
1	~4-6 hours	Phase 0 (clean build needed)	Test audit + bluff hunt in parallel
2	~3-4 hours	Phase 1 (must know which tests are real)	Task 2.3 (API verify) overlaps
3	~4-8 hours	Phase 2, Containers submodule, operator hardware	Matrix cells in parallel
4	~2-3 hours	Phase 1 (fresh audit after fixes)	Bluff hunt independent of phase 3
5	~1-2 hours	Phases 1-4, operator device/hardware	Manual steps (operator)
6	~2-3 hours	Phase 5	Builds in parallel (API + Android)
Total	**~18-29 hours**		

Parallelizable groups:

- Phase 0.1 (auth test fix) is independent of 0.2-0.9 (submodule/git cleanup)
- Phase 1 tasks can run in parallel across multiple agents
- Phase 3 matrix cells are embarrassingly parallel
- Phase 6.1+6.2 (Go build) and 6.3+6.4 (Android build) can run simultaneously

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Fallback

If operator hardware/emulator access is unavailable:

- **Phase 3** is replaced by: "Execute C17-C22 on any available device, document gaps for emulator matrix"
- **Phase 5.2** (real-device smoke test) is replaced by: "Document which steps require operator hands-on"
- **Phase 6.6-6.7** (tag cutting) uses `--no-evidence-required` as a dry run, with the caveat that tagging is NOT complete without the evidence pack